

Models of the motor system

Fall 2021

Opher Donchin

Part 1

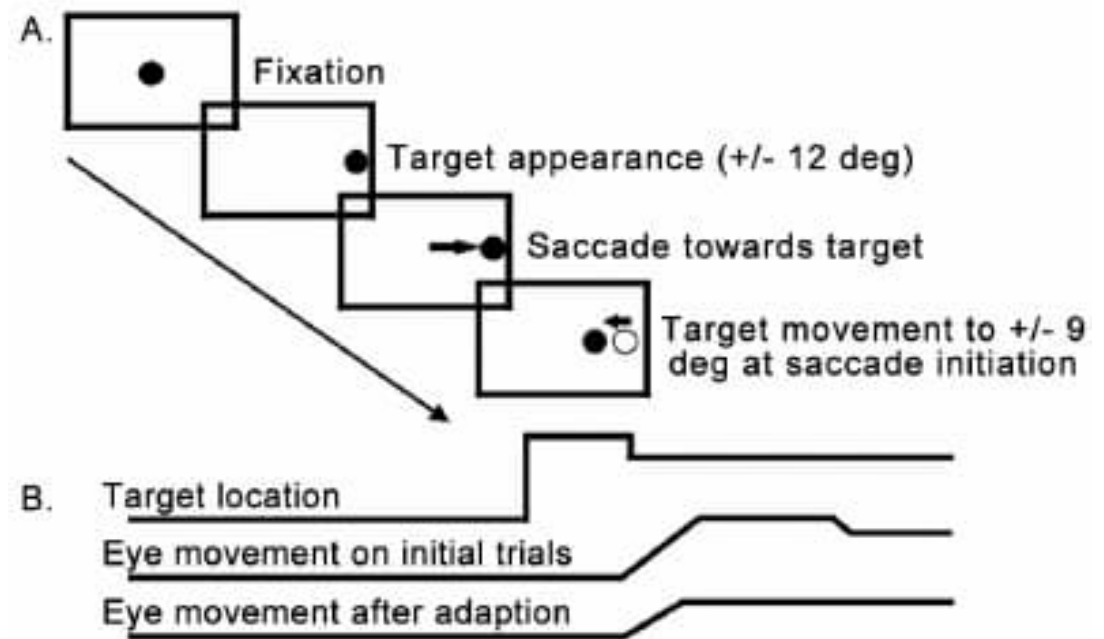
- Limitations of the error based model

Types of eye movements

- Fixation
 - Vestibulo-ocular reflex
 - Optokinetic reflex
 - Micro-saccades
- Tracking
 - Smooth pursuit
- Gaze shift
 - Saccades

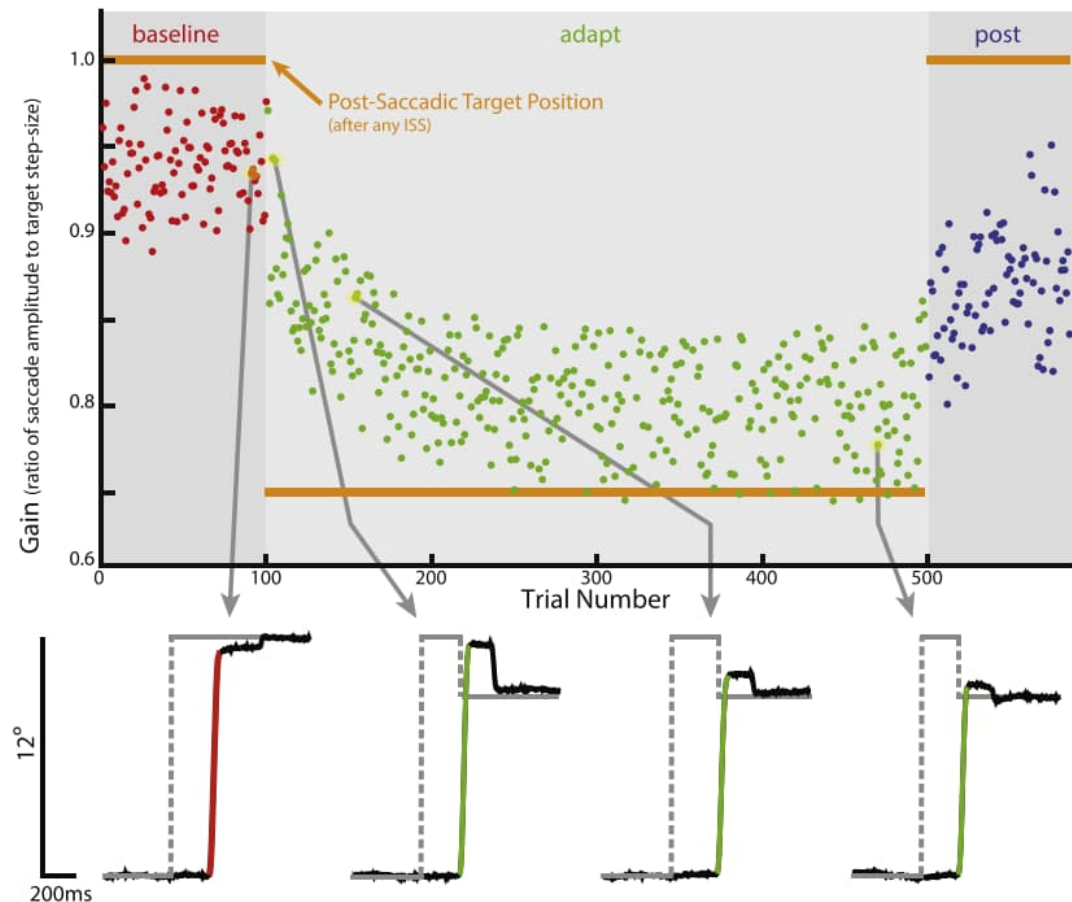
Saccade adaptation

- Saccade blindness
- Target jump during saccade



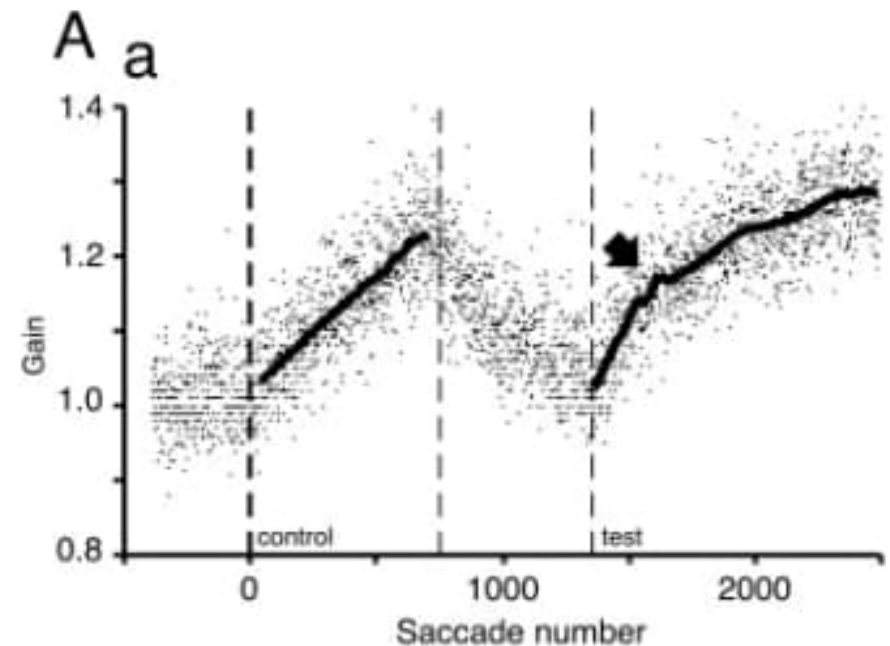
Saccade adaptation

- Saccade blindness
- Target jump during saccade
- Adaptation



Savings in saccade adaptation

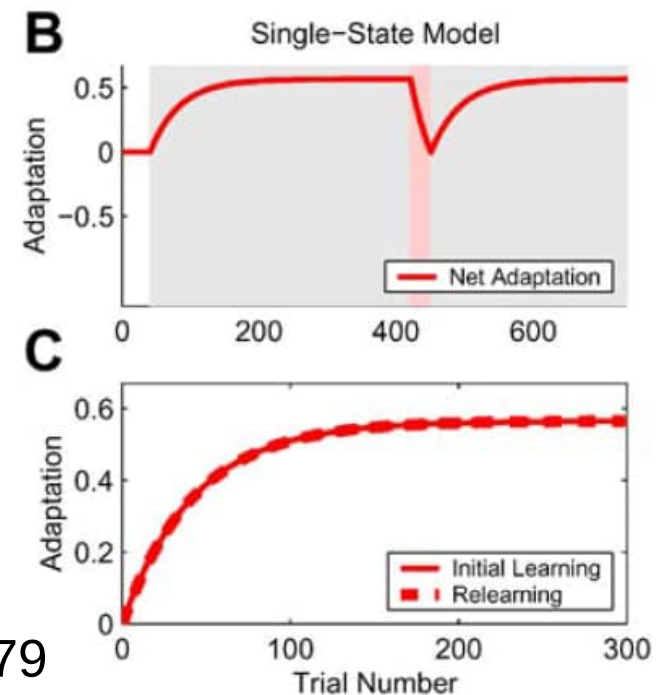
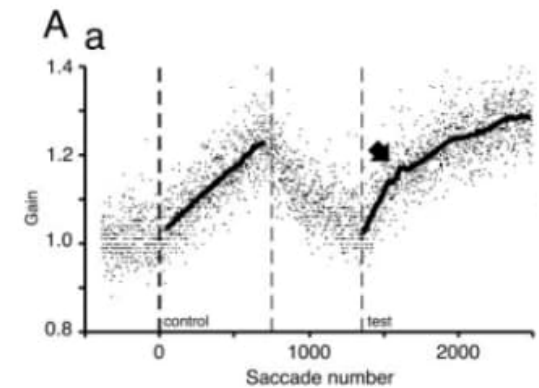
- More rapid re-learning
- Widespread learning phenomenon



Savings can't be explained by one state model

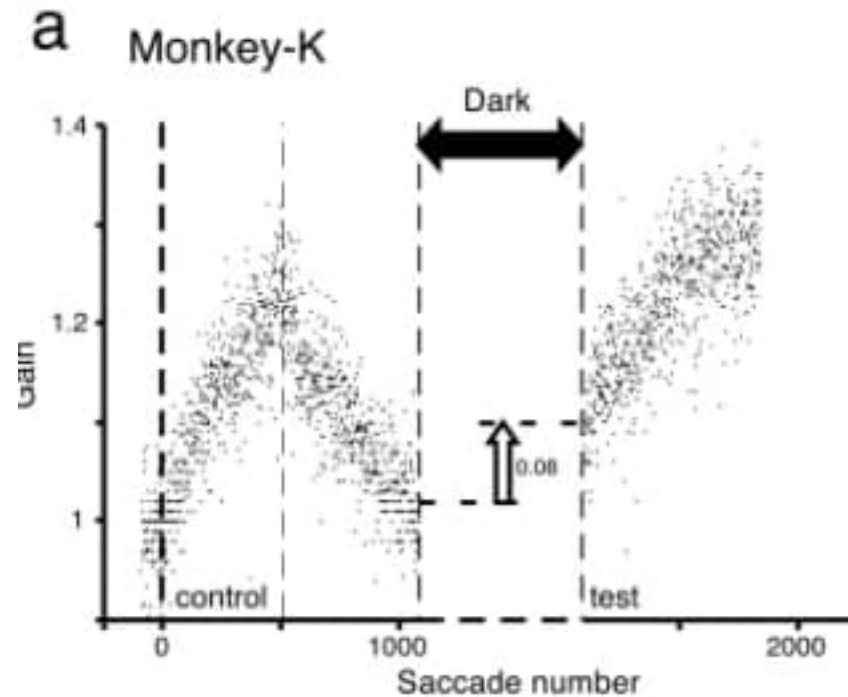
- Once the state is gone, it's gone

$$\begin{cases} x_{i+1} = Ax_i - Be_i + \eta_i \\ e_i = x_i - p_i + \varepsilon_i \end{cases}$$



Spontaneous recovery

- Unlearning doesn't wipe out learning



Part 2

- The two-state model

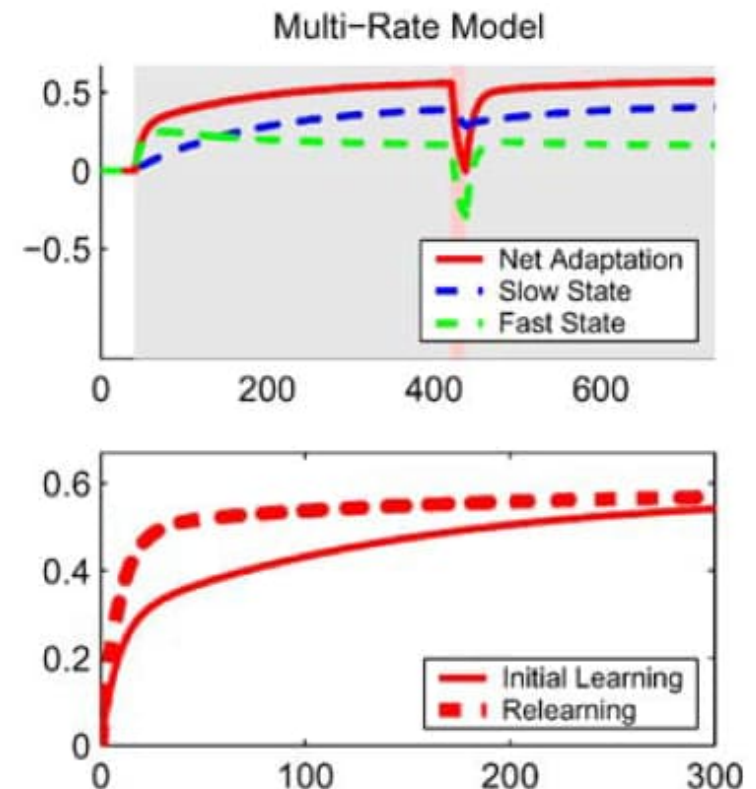
The two-state model

- Two time constants
 - Fast: fast learning + fast forgetting
 - Slow: slow learning + slow forgetting

$$x_{f,n+1} = A_f x_{f,n} + B_f e_n$$

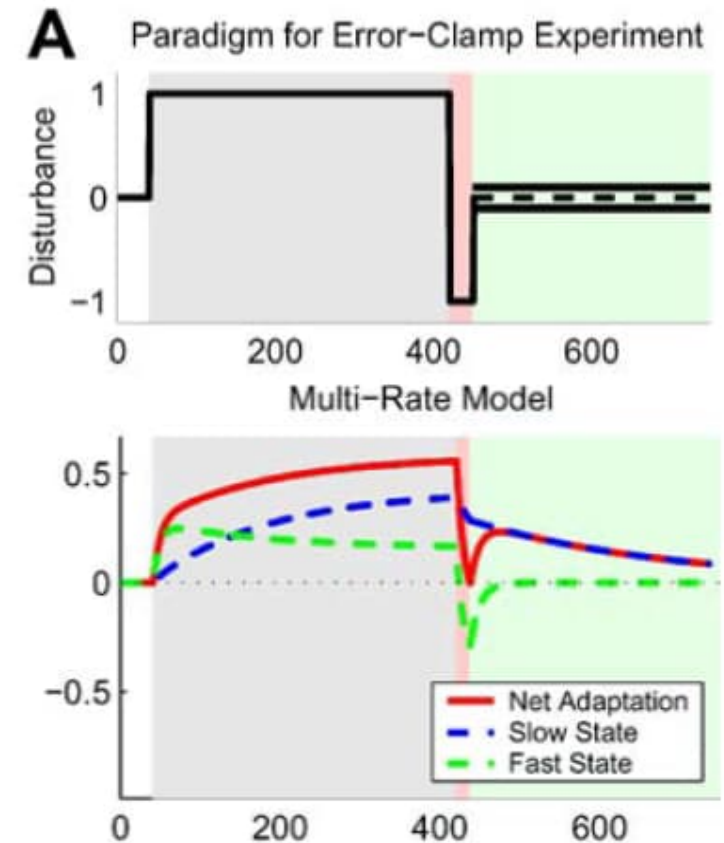
$$x_{s,n+1} = A_s x_{s,n} + B_s e_n$$

$$x_n = x_{f,n} + x_{s,n}$$

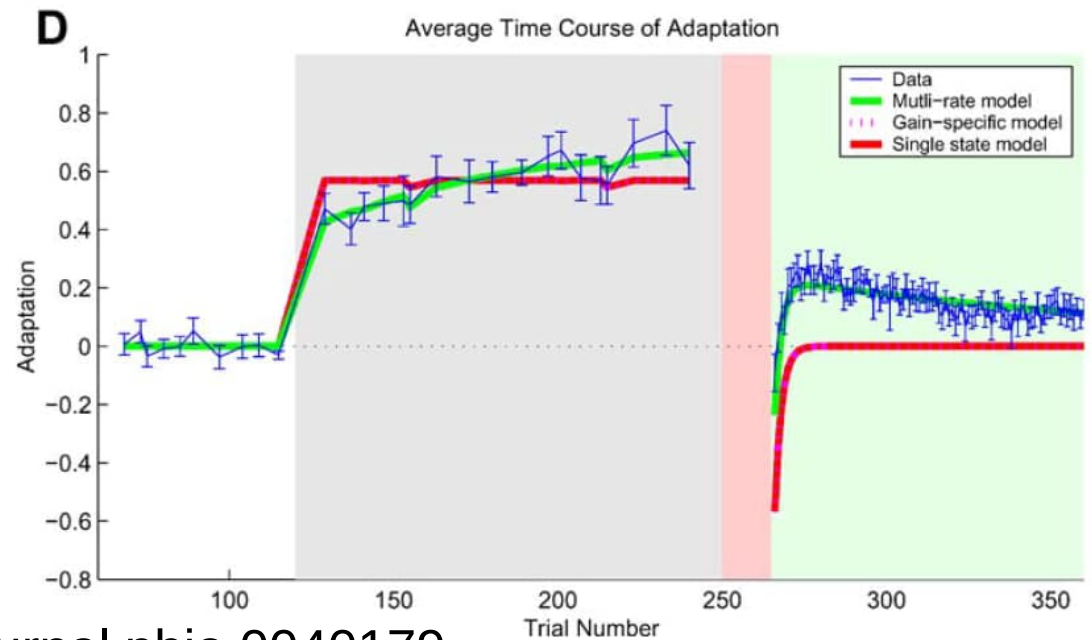
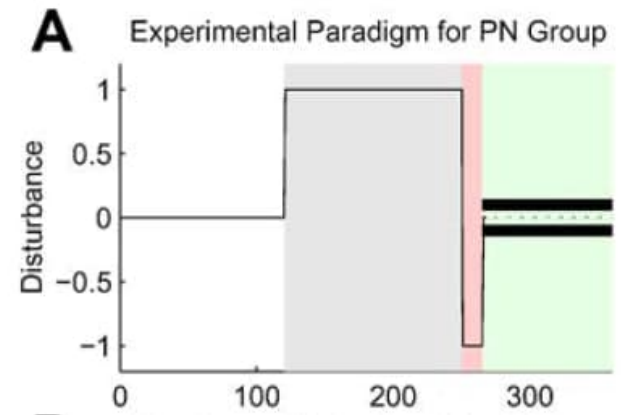


Two state model explains spontaneous recovery

- Error



Experimental verification

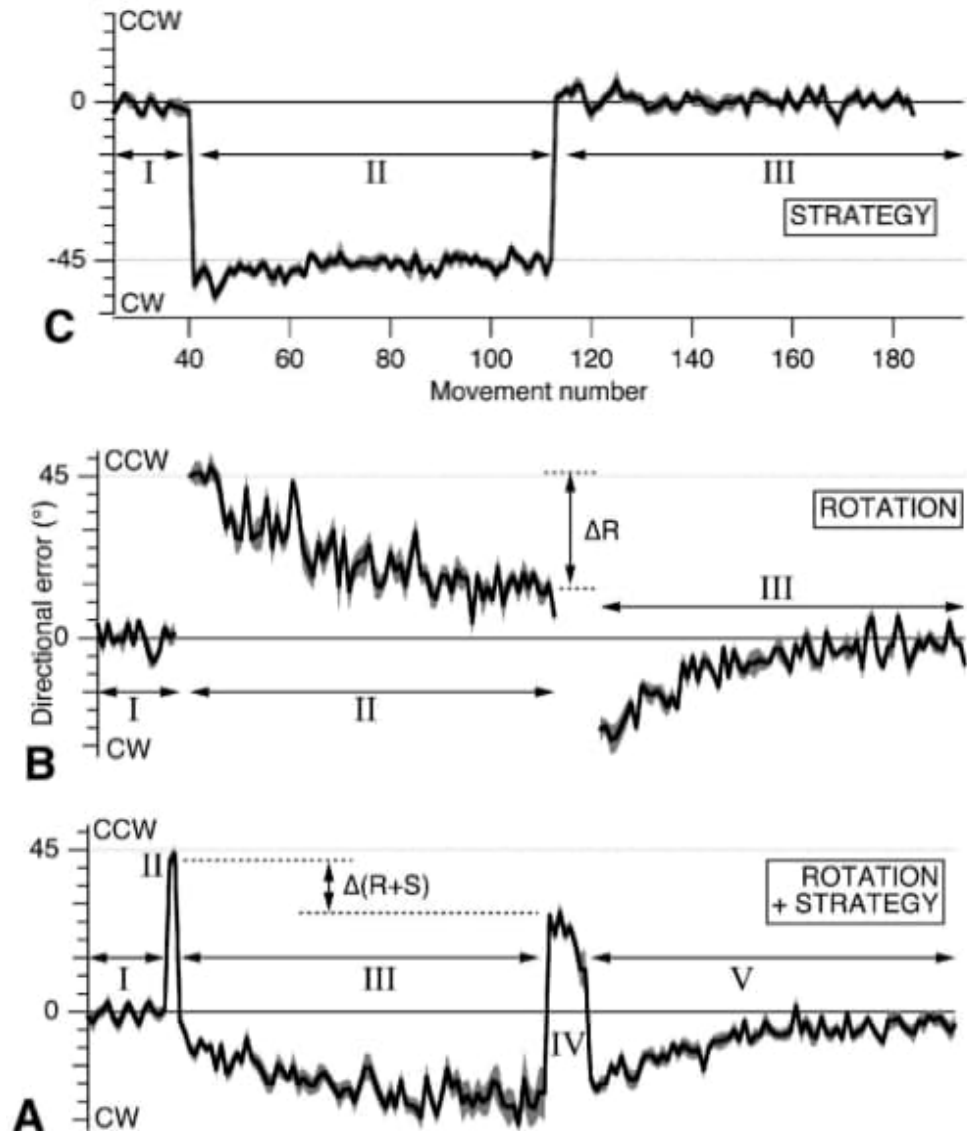


Part 3

- Explicit and implicit adaptation

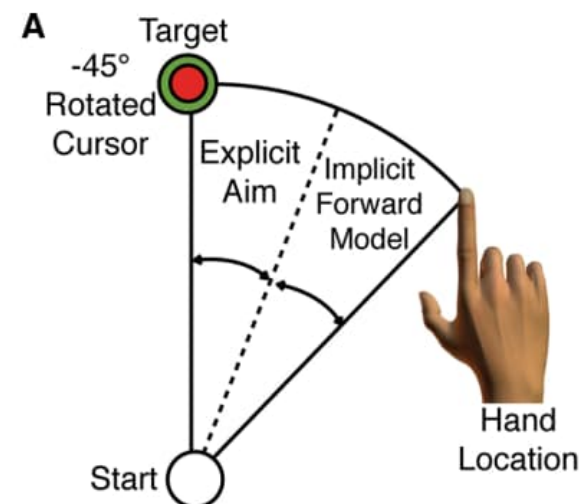
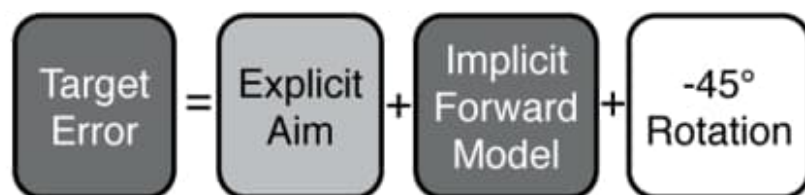
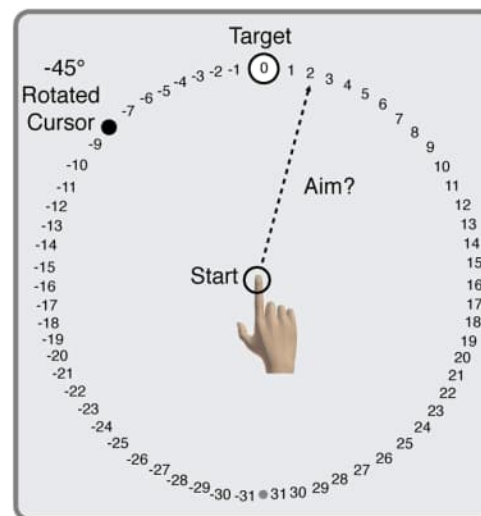
Interaction of adaptation and strategy

- Strategy alone
 - People can move where we ask them
- Adaptation alone
 - People can adapt
- Interaction
 - Adaptation precedes **despite** strategy

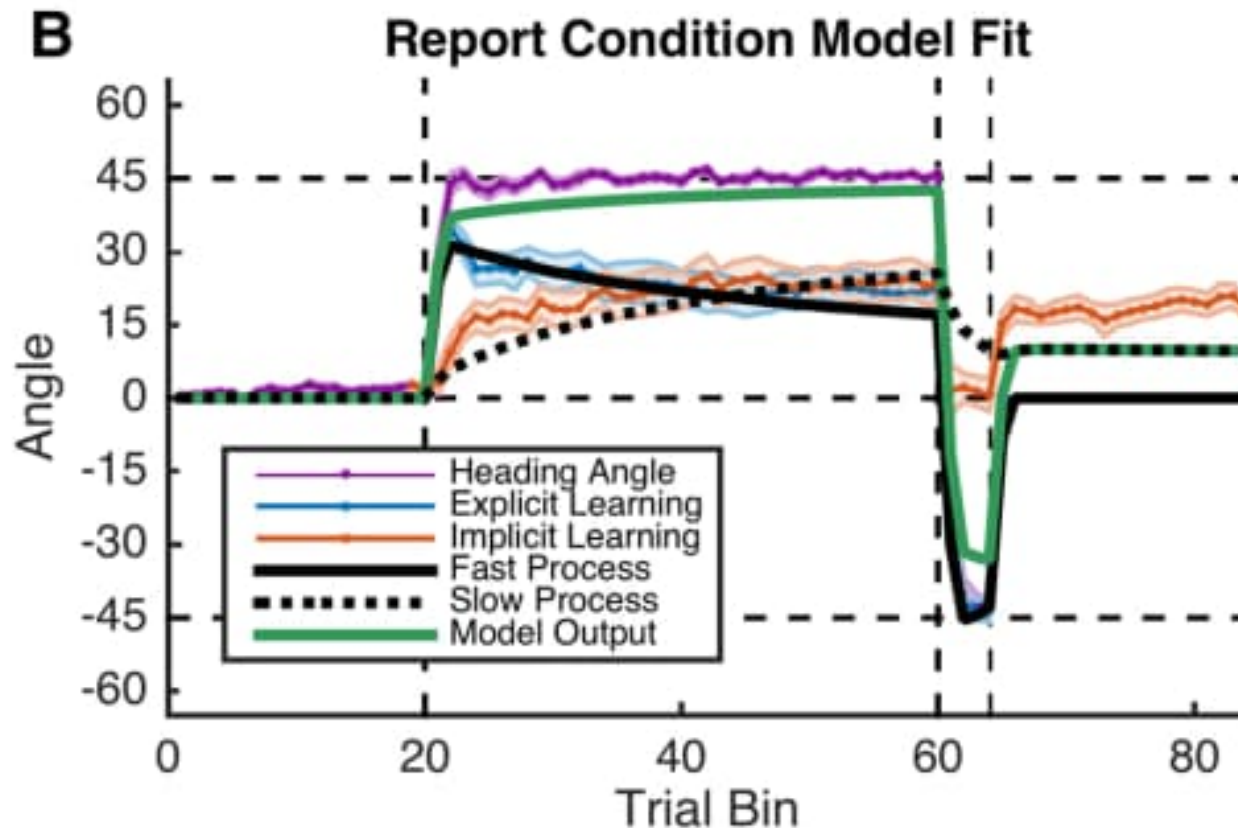


Measuring explicit intention

- Just ask
 - Where are you aiming?
 - The rest is implicit

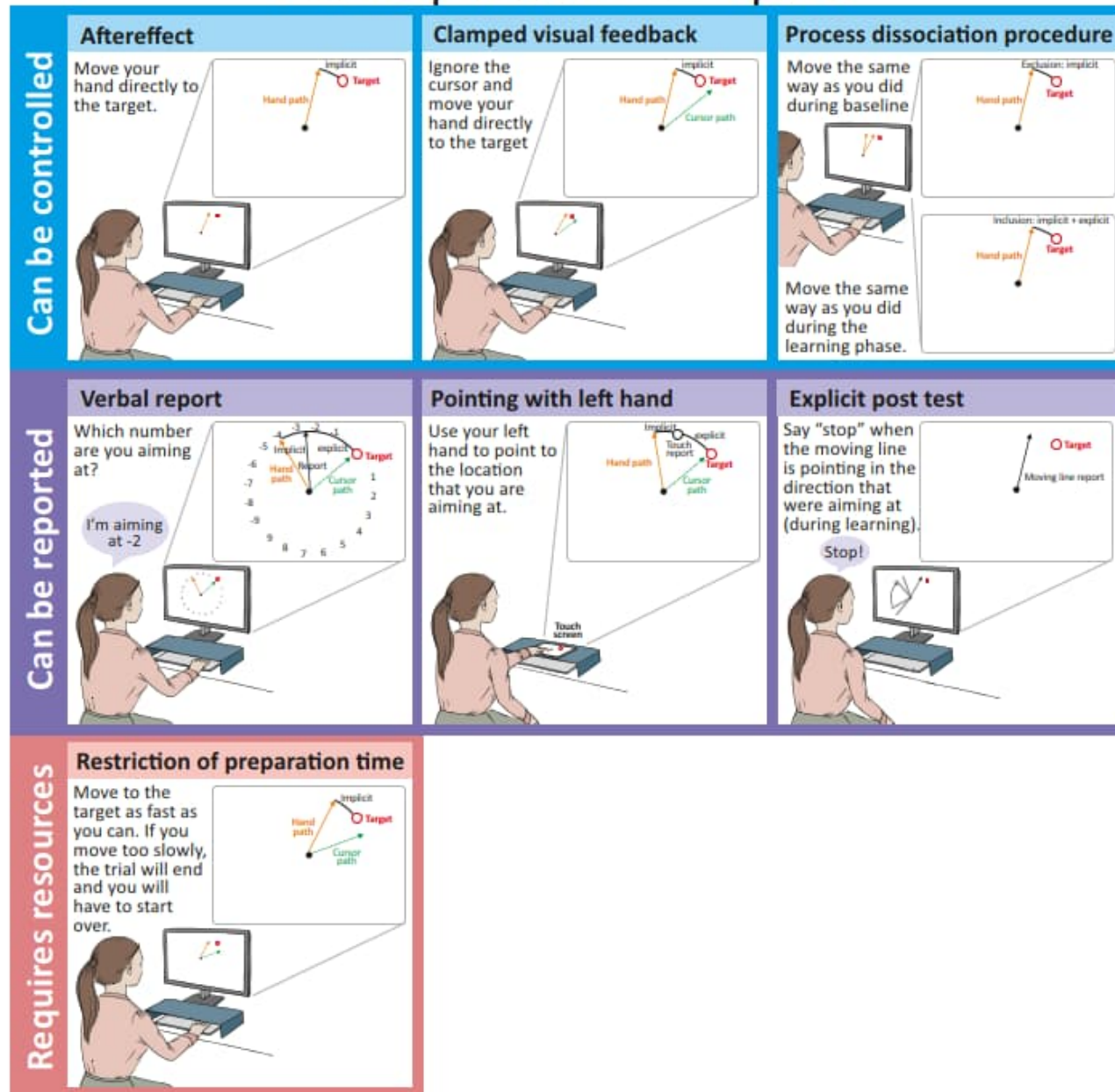


Explicit and implicit match fast and slow



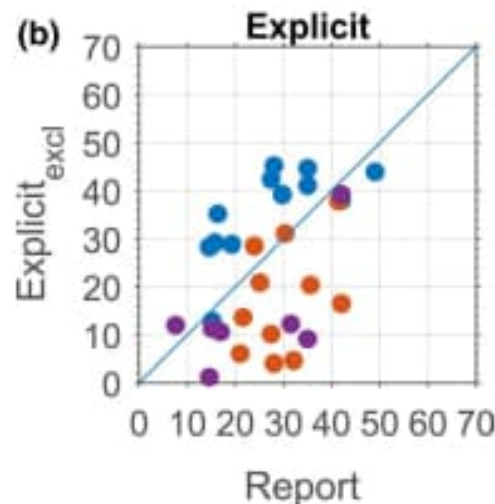
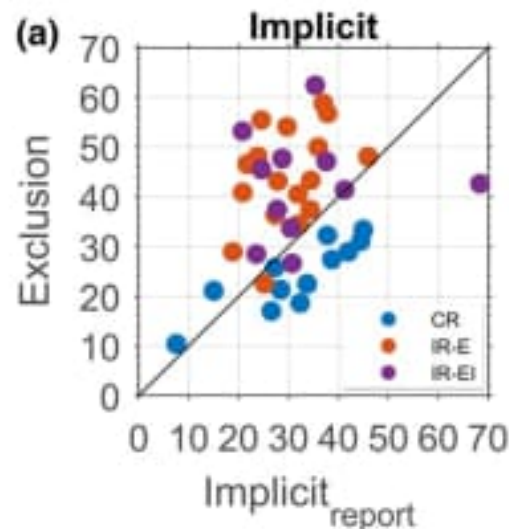
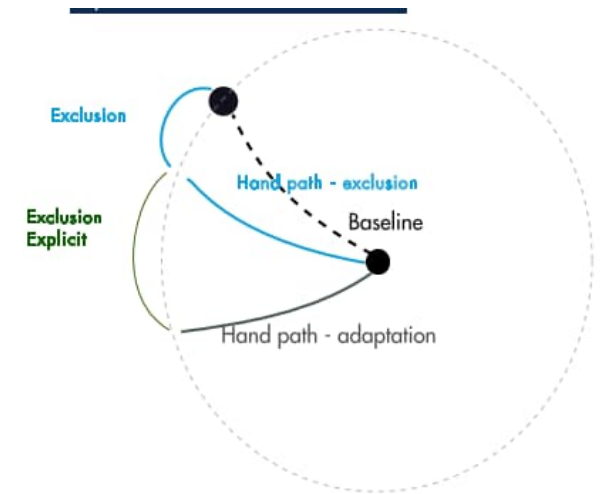
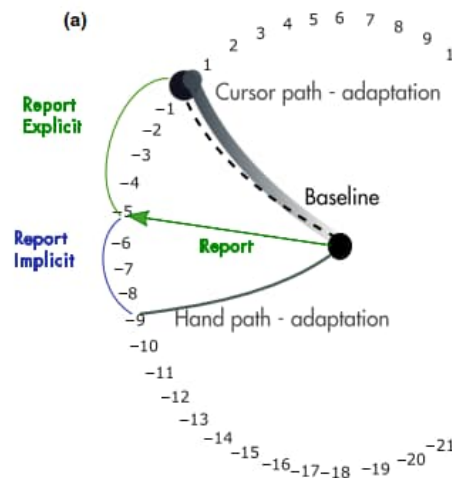
Many ways to measure intention

Assumption about explicit



The measure matters

- Consistent reporting
 - Report shows more explicit
- Intermittent reporting
 - Exclusion show more explicit



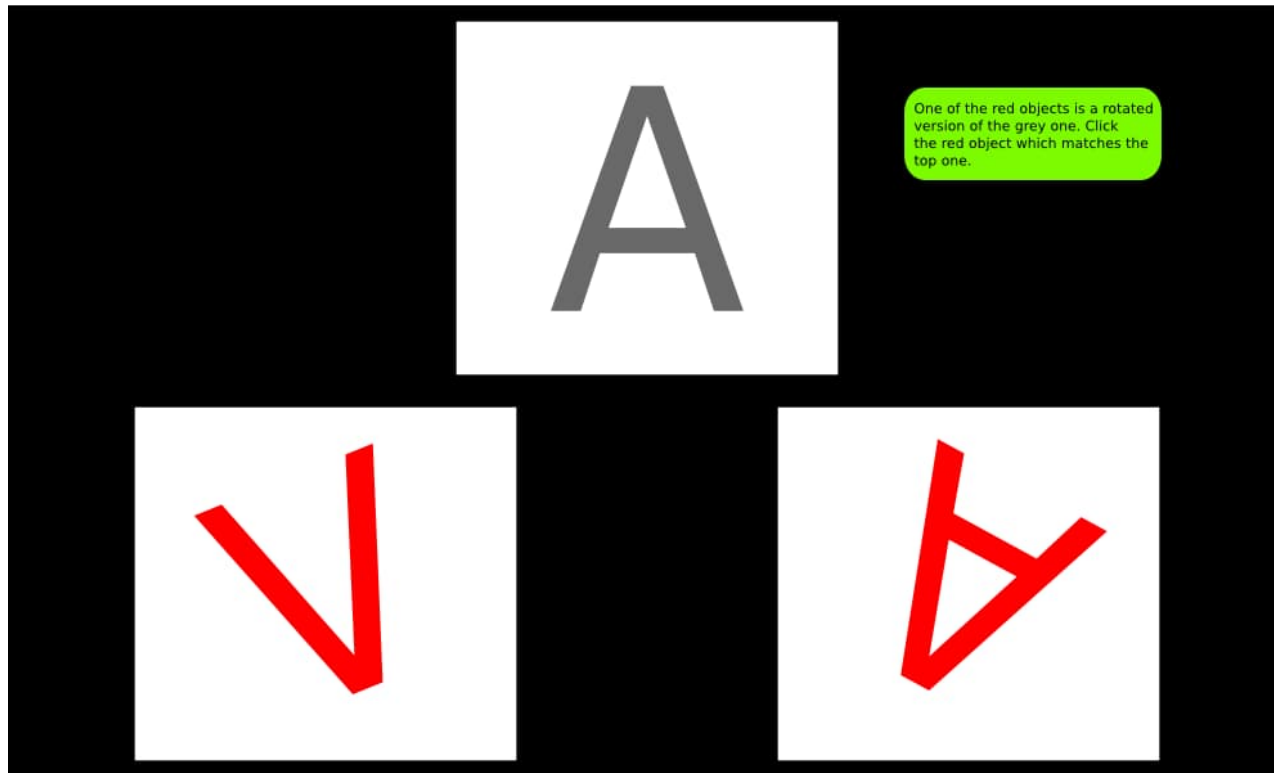
CR: consistent reporting
IR: Intermittent reporting

Part 4

- Sub-components

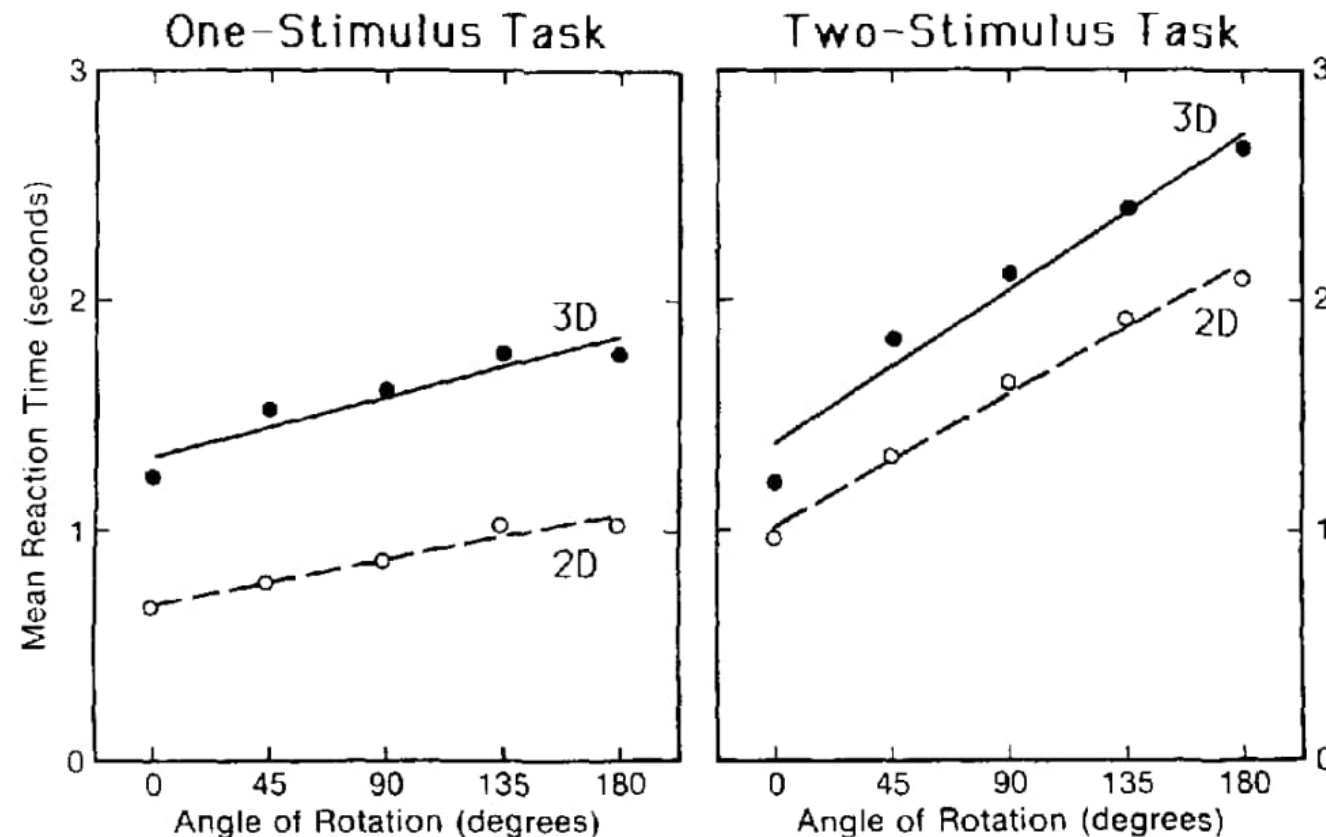
Mental rotation tasks

- Try it
 - https://www.psychtoolkit.org/lessons/experiment_mentalrotation.html



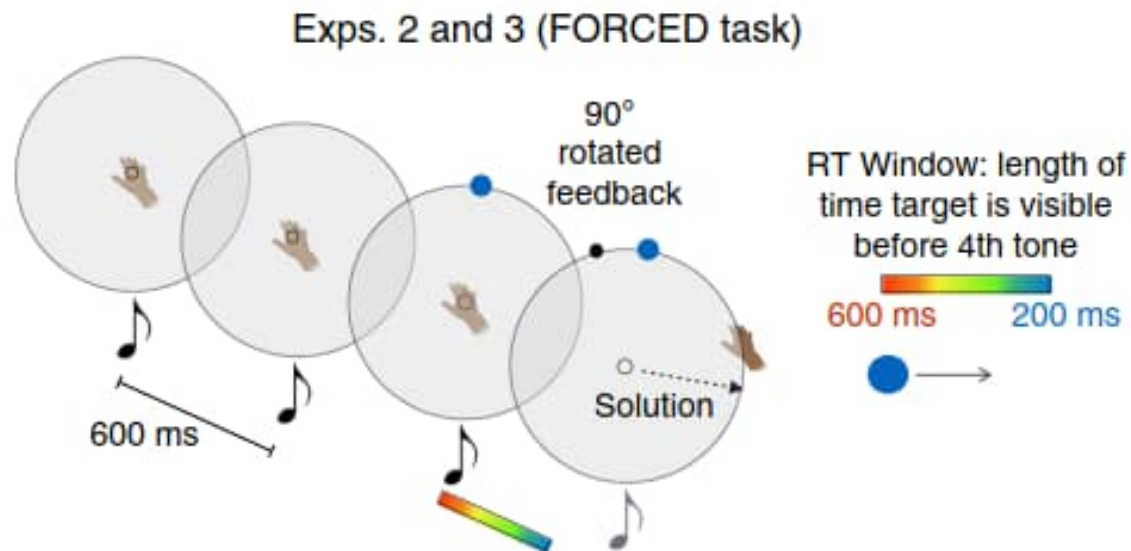
Reaction time scales with rotation distance

- We know you are performing mental rotation because it takes you time to rotate



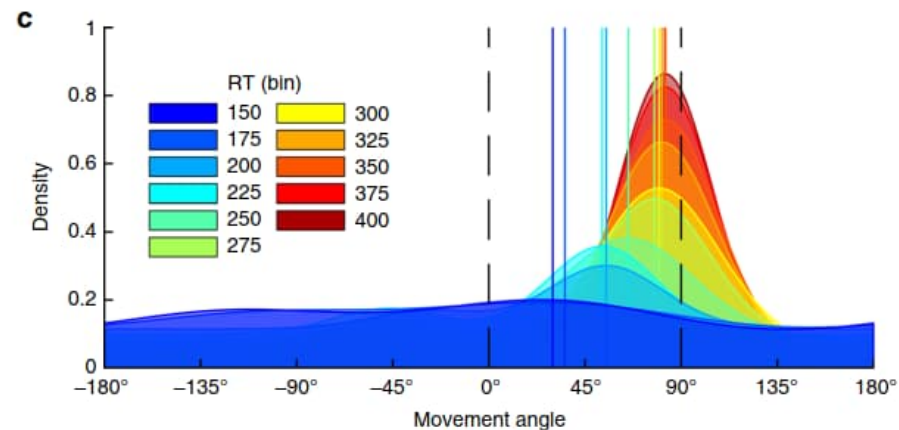
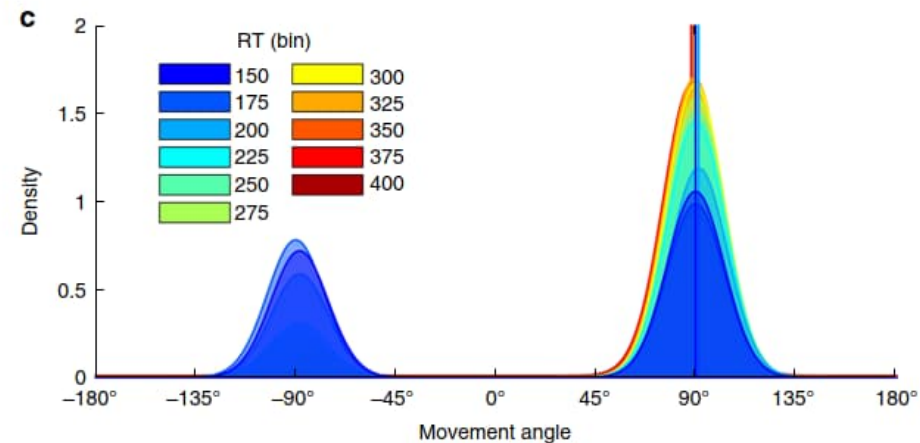
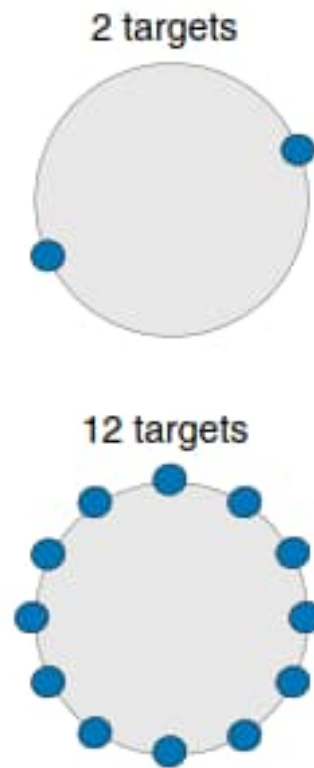
Mental rotation in adaptation

- Use tones to force a reaction time
- How will adaptation be affected by reaction time?



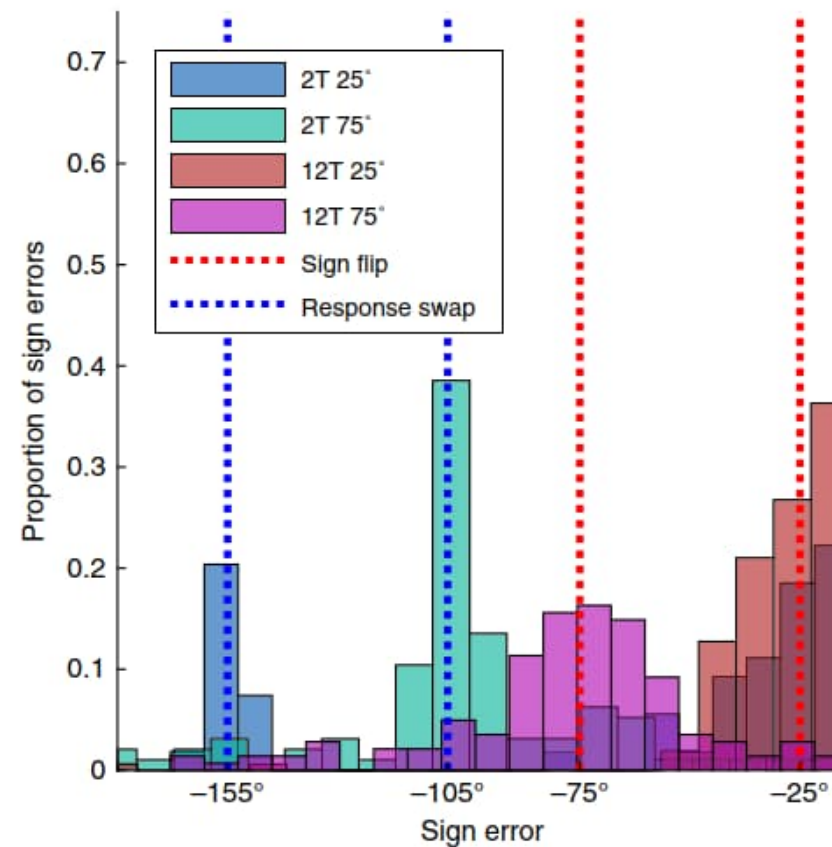
Mental rotation strategy

- Multiple targets forces a rotation strategy



Caching strategy

- With 2 targets
 - Mistakes of going to the wrong target
- With 12 targets
 - Rotating the wrong way

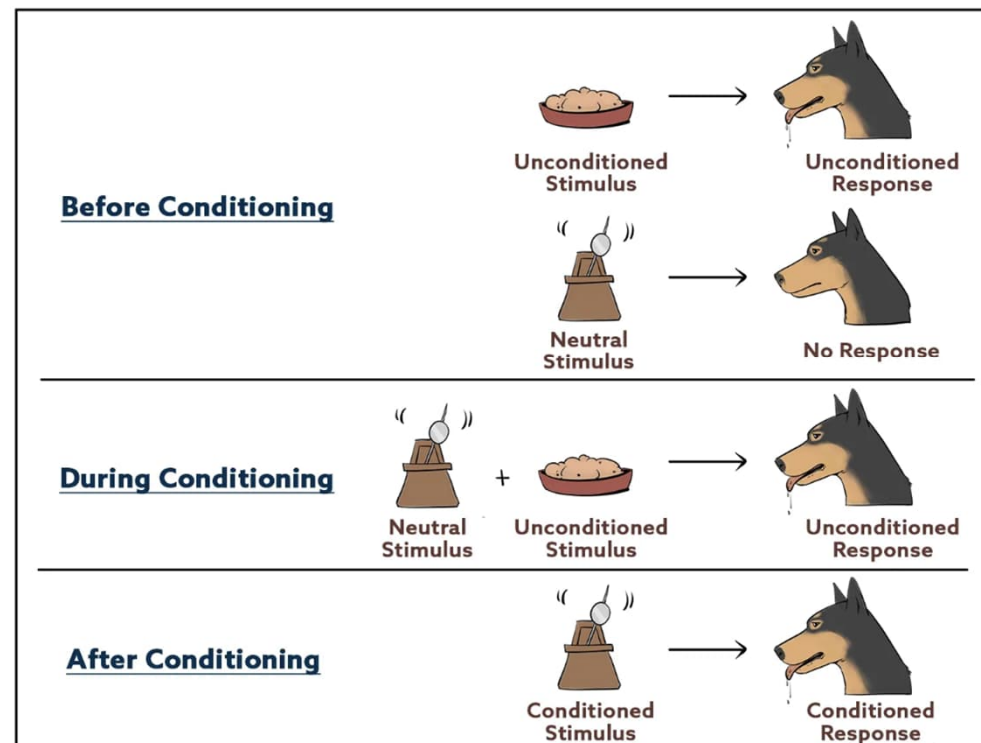


Associative learning model

Associative learning

- How do we associate cues with appropriate responses

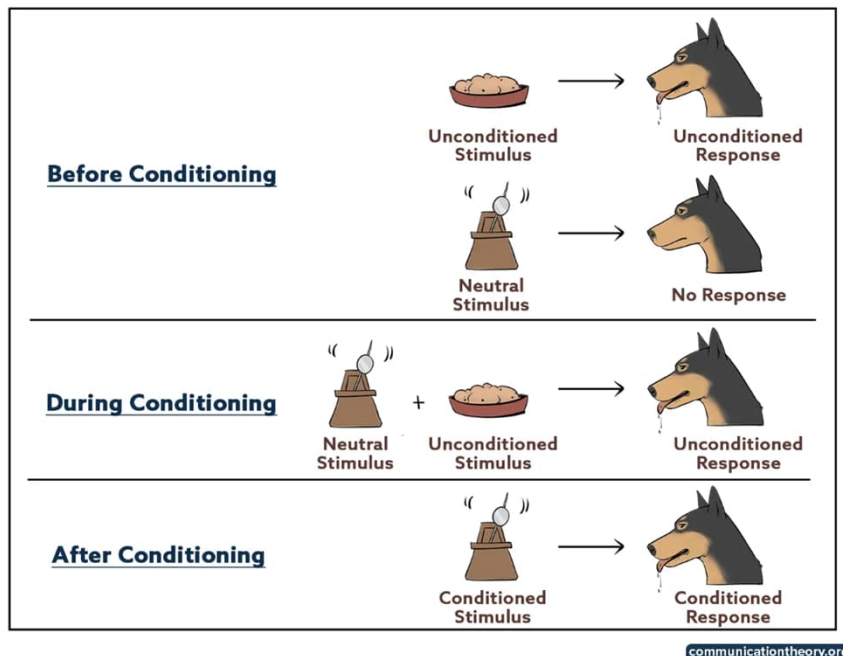
CLASSICAL CONDITIONING THEORY



Rescorla-Wagner model

- How do we associate cues with appropriate responses

CLASSICAL CONDITIONING THEORY



$$V^{[n]} = V^{[n-1]} + \alpha\beta SPE^{[n-1]}$$

$$SPE^{[n-1]} = \lambda - V^{[n-1]}$$

λ : Value of US

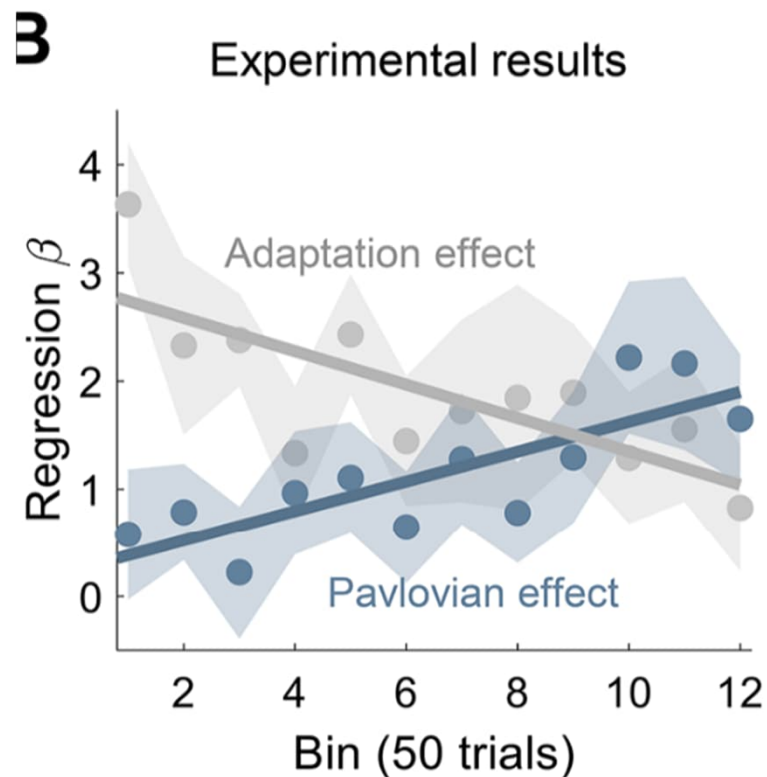
V : value of CS

SPE : Sensory prediction error

α : Salience of CS

β : Learning rate

Success of Rescorla Wagner



$$V^{[n]} = V^{[n-1]} + \alpha\beta SPE^{[n-1]}$$

$$SPE^{[n-1]} = \lambda - V^{[n-1]}$$

Adaptation effect: Change in CS+
Pavlovian effect: CS+ - CS-

More conditioning experiments

- Compound cues
 - If you receive both a sound and a light
 - Each receives part of the credit
- Prospective Interference
 - If you learn that a light causes the perturbation
 - You block later learning of a sound

A two time constant model

- Fast
 - Responsibility
- Slow
 - Association

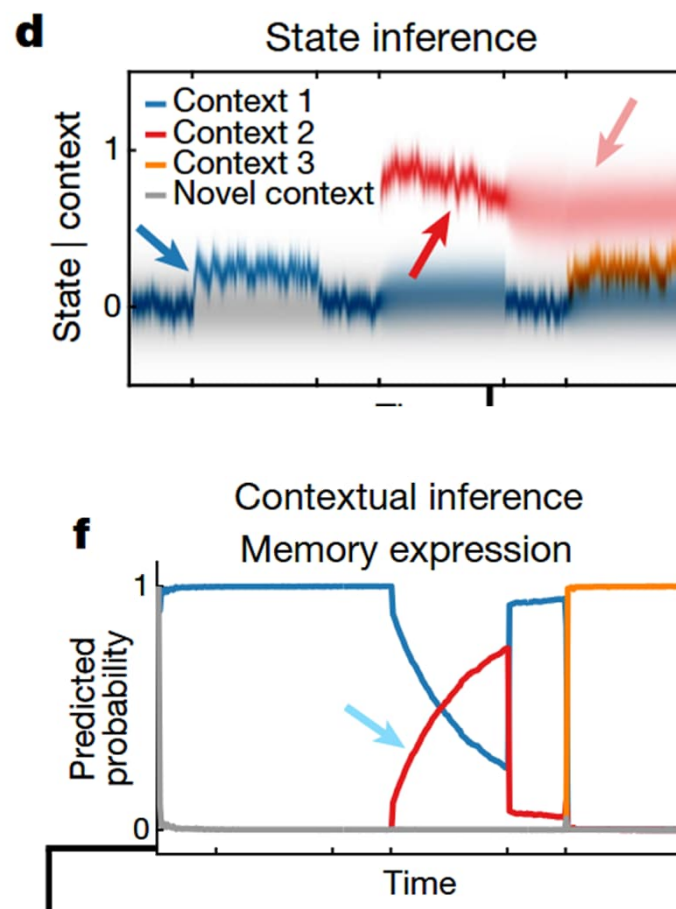
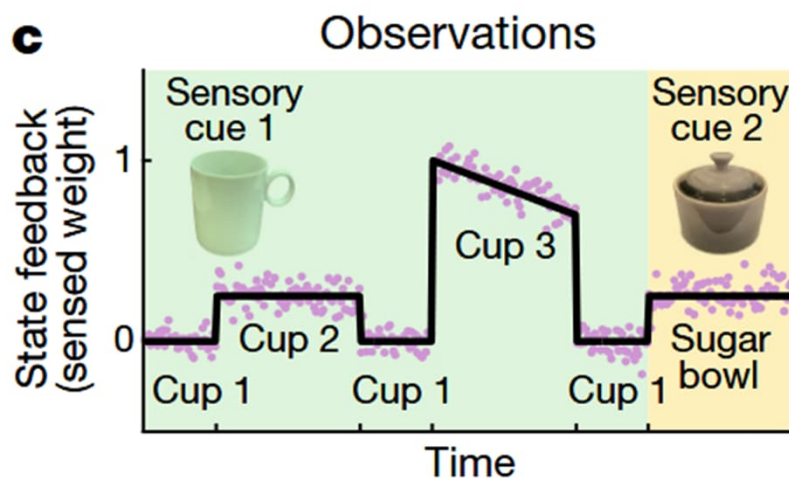
The COIN model

- Contextual inference model

Inferring context is important

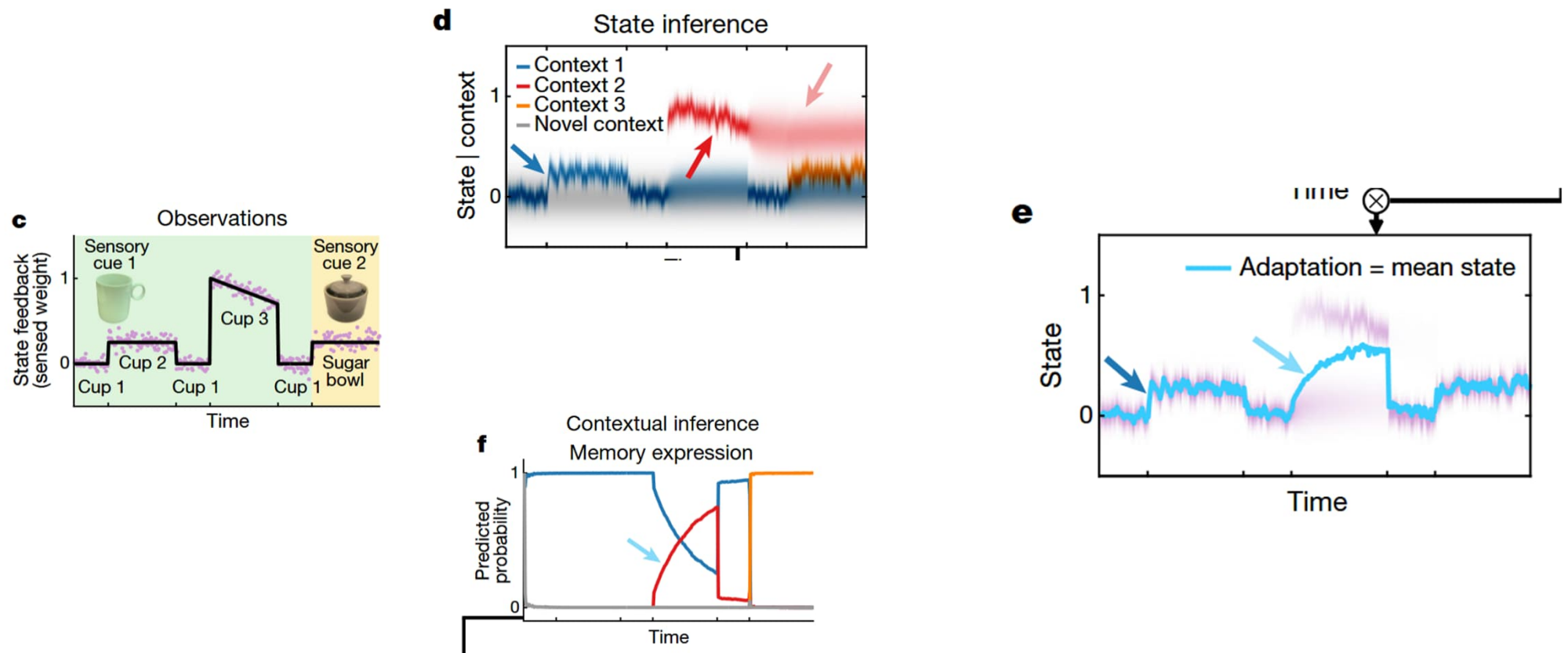
New context is inferred if:

- The error is large
 - Cup 3
- There is a cue
 - Sugar bowl



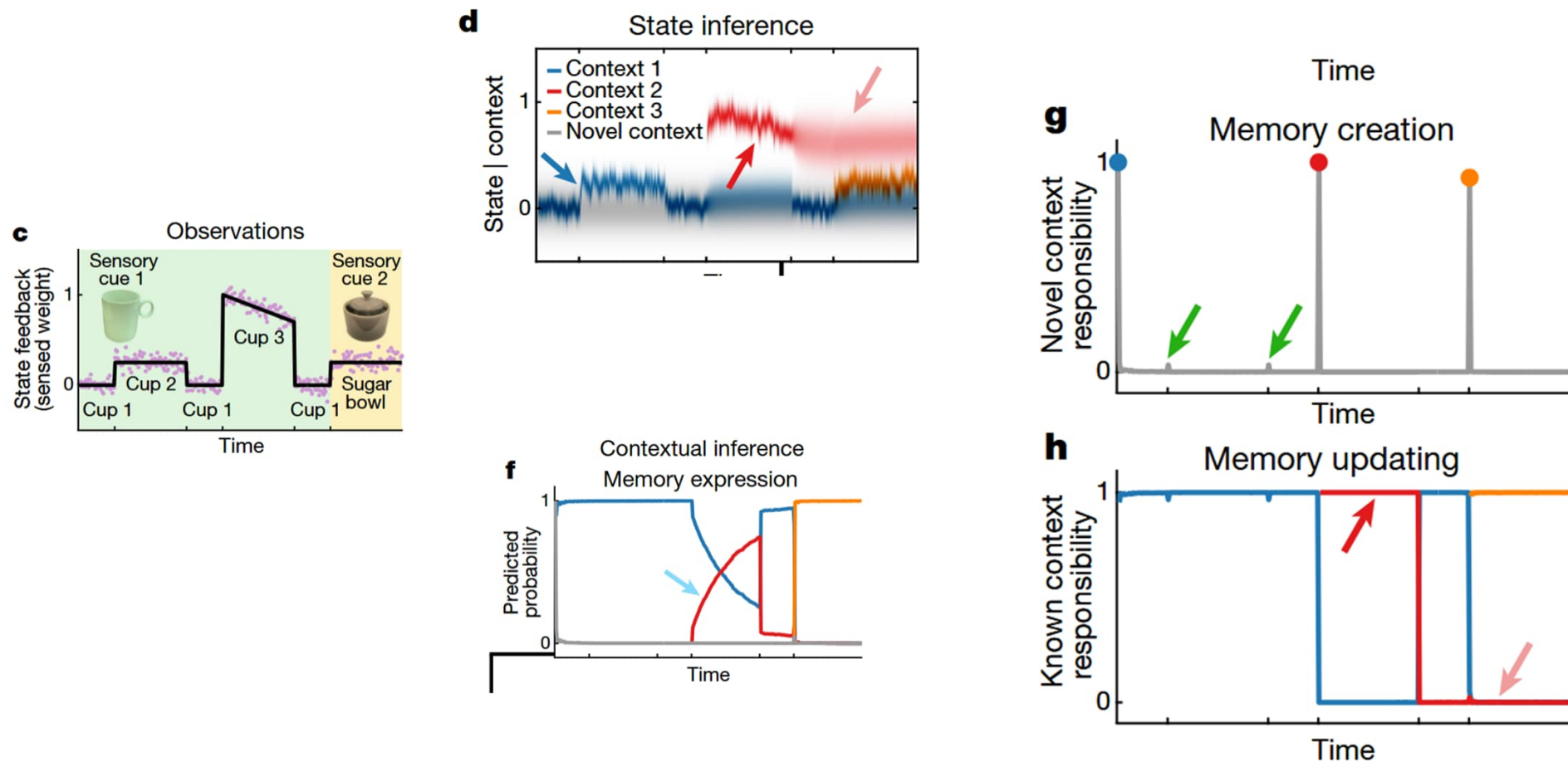
Contexts combine according to probability

Kalman filtering applied to action not perception



Novelty leads to context creation

The Chinese restaurant process



Still two time constants

- Fast
 - Context switching
- Slow
 - Drift

Looking forward

- Two more lectures
- Homework 3 handed out today
- Projects on the last week of class

Projects

- Choice of 4 papers
 - Each student a different paper
- Template of presentation
 - Slide titles with questions
 - You fill in the figures and equations for answers
 - No additional text on slides
 - I will share a grading rubric